Design and control of a biologically inspired shoulder joint

Abstract. In the field of robotics, human-centered, we need flexible, robust, agile robots that have the natural human mechanisms in place. In order to achieve these robots, human motor systems must be analyzed and abstracted, from the mechanical level, to the behavioral and cognitive level. In this paper I will present abstracted the human shoulder wrist, seen as a modified Stuart platform, with a platform supported on a pivot and driven by four actuating motors. The elements of resistance imitate the shoulder blade and the humerus, and those of action and control of the movements represent the four main muscles of the shoulder. They have the shape close to the natural shape of the human bones, and the elements of action and control, which take over the functions of the muscles, have the location and the points of connection with the elements of resistance as well as the natural connections between bones and muscles. The connections between the natural muscles and the bones are made using the tendons. These flexible links, coupled with the open shape of the shoulder wrist, drive the humerus in a 360 degree rotational motion. To allow a "left-up-right-down-left" circular motion, we introduced the "tendon" that has a cardan coupling, on the side of the "muscles" and a sphere on the side of Humerus.

Keywords: robot control, bio inspired, shoulder joint

1 Introduction

In most cases, with the help of humanoid robots, we try to imitate the natural movements of humans, but the mechanisms used are different, anatomically speaking, from those of humans. However, efforts are being made to control the trajectory of the robot members so that it is as smooth as possible [1]. Obviously, the robots used in different industrial environments have the specific movements of the work submitted, movements reduced to the essential, and the humanoid robots, which have implemented the same mechanisms, have the same movement patterns. But if a robot or just one arm of it interact with the human or be part of an exoskeleton, attached to a part of the human, then it should copy not only the shape but also the internal structure of the human body, so that human-machine interaction can as natural as possible.

Two problems arise in the construction of biologically inspired humanoid robots [2], the first refers to the creation of the skeleton and the muscles that create a unitary whole from the skeleton and with which to control parts of the skeleton, and the second refers to the abstracting of the component elements. to make them easier and easier to control [3].

Humanoid robots are those robots that take not only the human form, but also the structure of the skeleton and biological functions, which allow the perception of the environment, at the sensory level, and the appropriate response of the system to the action of the environment on it. The natural movements, specific to the human being, can be realized by combining a realistic skeleton driven by artificial muscles composed of actuating motors and elastic links or with several joints. For the control of artificial muscles, the actuating motors take on the functions of the motoneurons, which naturally control the muscle fibers, thus creating a motor unit [2]. The motor unit of an artificial muscle has two phases, one of contraction and one of relaxation. For this the motor units must be paired, working in tandem, when one contracts the other relaxes, leading the arm in the direction of contraction. By combining the movements of two neighboring actuating motors (left-up, left-down, right-up and right-down), the contraction replaces the movement of a "muscle" that would make a diagonal movement.

2 State of the art

The first steps to control this complex robot were taken by Ivo Boblan and the team, in 2004, when they created the ZAR3 robotic arm [4]. The control of the movement of the humerus in the wrist is done by cables operated with three pairs of actuator motors for each axis. Steffen Wittmeier and Alois Knoll [1] present in 2010 the anthropomimetic robotic arm ECCEROBOT, which tries to imitate the structure and functionality of a human arm, using 11 actuating motors and elastic strings for artificial muscles. Later Steffen Wittmeier joined Owen Holland, Rob Knight [5], to create the next ECCE1 anthropomimetic robot, this time with chest and two arms, with 43 actuators, with related sensors and control systems mounted in thorax. Bruno Dehez and Julien Sapin present the ShouldeRO exoskeleton arm, composed of 3 Cardan-type connectors, for displacement on the three planes, sagittal, horizontal and frontal [6].

3 About imitation and abstracting the idea

The imitation and abstraction of the forms and functionalities of the natural structures consists in taking over the model of the natural body shape, which is to be copied, and the functions of the component parts, and then adapting them to an electromechanical model. Imitation of the form as a whole, taking over the component elements, of the form, of their number and order in the natural organism. Imitating the functionality of the elements that produce the movement, of the muscles, for example, which have effect only at the time of their contraction, bind to bones at certain points, at certain distances from the ends of the bones, and together with the wrists form levers, to use at least energy for maximum efficiency. From the total number of muscles that work the wrist, we have chosen only four muscles, paired two by two, connected by the shoulder blade and humerus at the points where the natural muscles are connected by the respective bones. The robot arm can be controlled by an operator using a controller, a joystik for example, with the help of a neural headset [7], with EMG signals [8] or with an augmented reality device, which has implemented eye-tracking technology [9], in case where the arm goes where the operator looks.

"Deltoid muscle", positioned at the top, paired with the lower muscle, "Teres Major". On the right is placed the "Infraspinatus muscle", paired with the one on the left, the

"Subscapularis muscle". Because "Deltoid muscle" has the main role of supporting the weight of the arm, the actuator motor that controls its movements, is oversized, in size and power, compared to the other actuator motors of the other "muscles". On the movable part, by the equivalent of the Humerus, are attached the sockets of the arm. To these sockets are connected the spherical ends of the flexible links, the equivalents of the tendons. These sockets are attached by Humerus from an angle of 30 degrees. Between the sockets of the arm and the shoulder blade are the two contact elements, the head of the humerus, abstracted in the form of a ball pivot and the glenoid cavity, in the form of a hemisphere (see Fig. 1). The sphere of the pivot is considered in contact with the hemisphere as long as the center of the sphere of the pivot coincides with the center of the hemisphere. The extremities of the range in which the "neck" of the pivot can move, until it reaches the hemisphere edge, form an angle of 130 degrees.

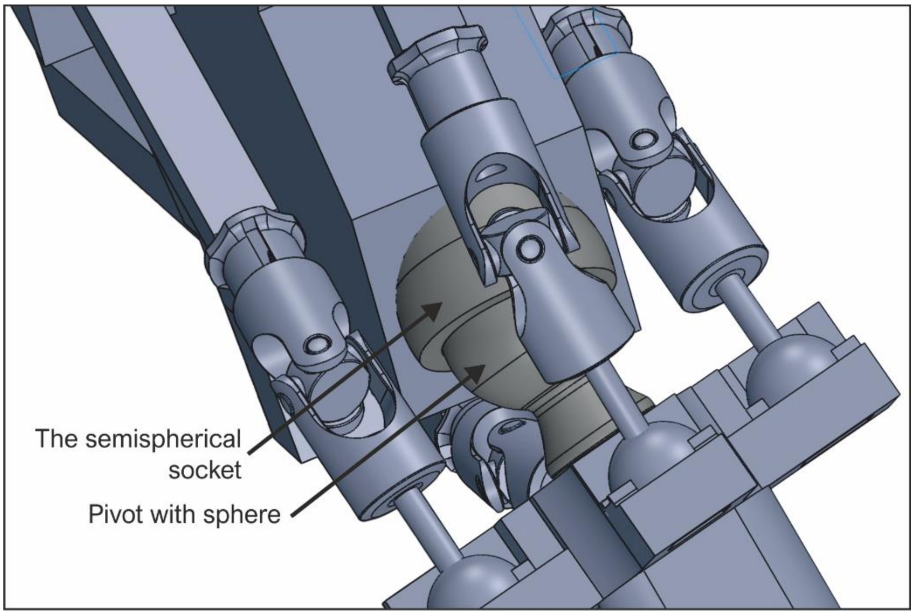

Fig. 1. Shoulder Wrist - The sphere pivot represents the head of the Humerus, and the hemispherical socket represents the Glenoid cavity on the shoulder blade.

4 Modeling and simulation

The proposed model of the shoulder joint mimics the natural model of the human shoulder, in that it has the supporting element, the shoulder blade, the actuators, the main muscles of the wrist, connected at the same points, on the shoulder blade and on the humerus, as the natural muscles and the humerus, as a movable element. These muscles work in tandem, two by two, on their contraction, so when a muscle contracts, its partner relaxes. In the presented model the rods actuated by the actuator motors represent the shoulder muscles, and the Cardan type links, for 360 degrees mobility, together with the contact sphere represent the muscle tendon (see Fig. 2 a and b).

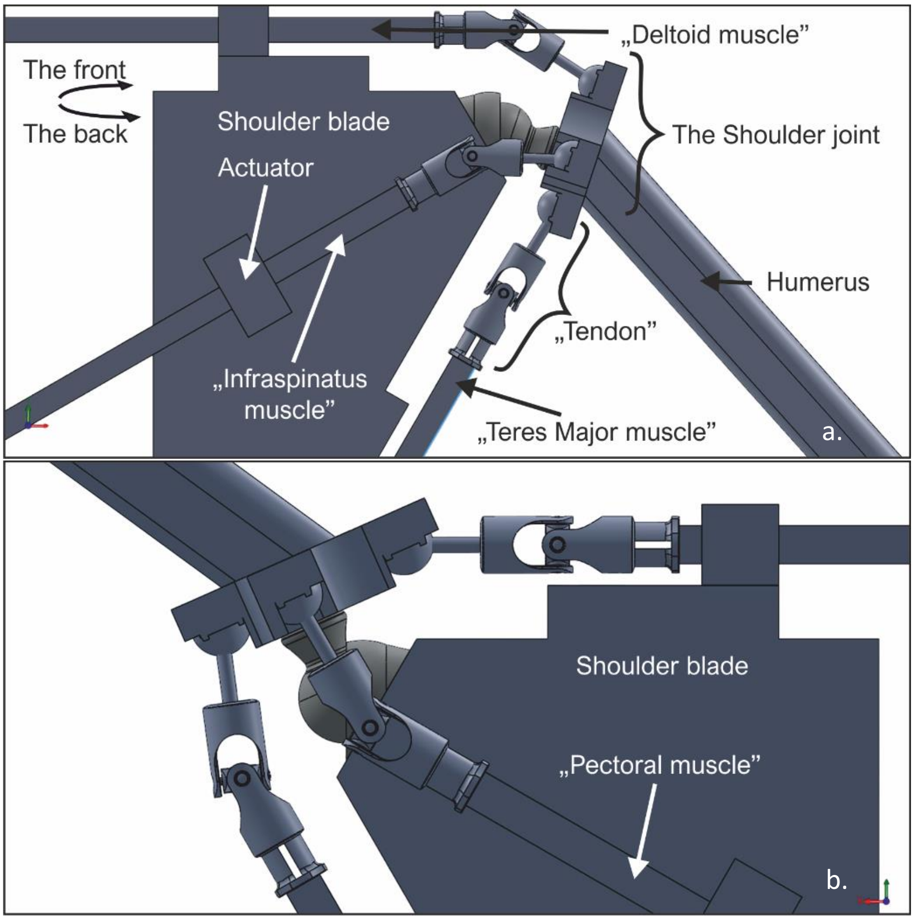

Fig. 2. "Muscles" and "tendons" of the wrist, a. The back and b. The front

When the "muscles will contract", ie the rods bolted by the actuator motors will pull the arm sockets towards the back, the sphere of the pivot will slide inside the hemisphere, leading the Humerus in the direction of the "contracted driver muscle". The actuator motors are continuously tensioned, so that the "muscles" will continuously pull on the sockets of the humerus, maintaining the integrity of the wrist, tensioning the joint. When it is desired to move the arm in a certain direction, one or two "muscles" will be used are "conducting muscles", because they will lead the arm in that direction. For example, if one wants to move the upper right arm, the conductive "muscles" will be "Infraspinatus" and "Deltoid", and "Subscapularis" and "Teres Major" will be the assistant "muscles". The movement of the conductive "muscles" is a synergistic movement, because the two "muscles" contract simultaneously, with the same tensions, leading the arm in the middle area between the two "muscles". When only one "muscle" is conductive, and his pair is assistant, the movement of the two is antagonistic, for one "contracts" and the other "relaxes". The arm will always move to the "muscle" part of the conductor (see Table 1.).

Table 1. Control and actuation elements

Muscles and their role (4 actuator motors + 4 flexible links)			
up on the shoulder blade	down on the shoulder blade	left (back) on the shoulder blade	right (front) on the shoulder blade
Deltoid	Teres major	Infraspinatus	Subscapularis
• as size and power is twice as large as the other actuators • pick up • tension	• tension • controls the lowering of the arm	• leads the arm to the left • conductor, together with the upper one, on the left-top diagonal • conductor, together with the lower one, on the left-down diagonal • tension	• leads the arm to the right • conductor, along with the upper one, on the right-up diagonal • conductor, together with the lower one, on the right-down diagonal • tension
• in synergy with the left or right actuators • antagonist with the lower one	• in synergy with the left or right actuators • antagonistic to the above	• in synergy with the top or bottom actuators • antagonist with the one on the left	• in synergy with the top or bottom actuators • antagonist with the one on the right

To simulate the shoulder joint design, the 3D model of the system was first imported into MatLab-Simulink-Simscape.

Figure 3.a shows the resulting MatLab model. In this you can see the couples and the elements. For actuating the robotic system, coupling actuators are added and a driving algorithm that takes into account the constructive mechanical constraints imposed on the arm design.

Figure 3.b shows a result of the simulation of the robotic arm, more precisely the horizontal stretched arm position. In order to bring the arm to any other position in its working space, the motors of the couple motors are realized.

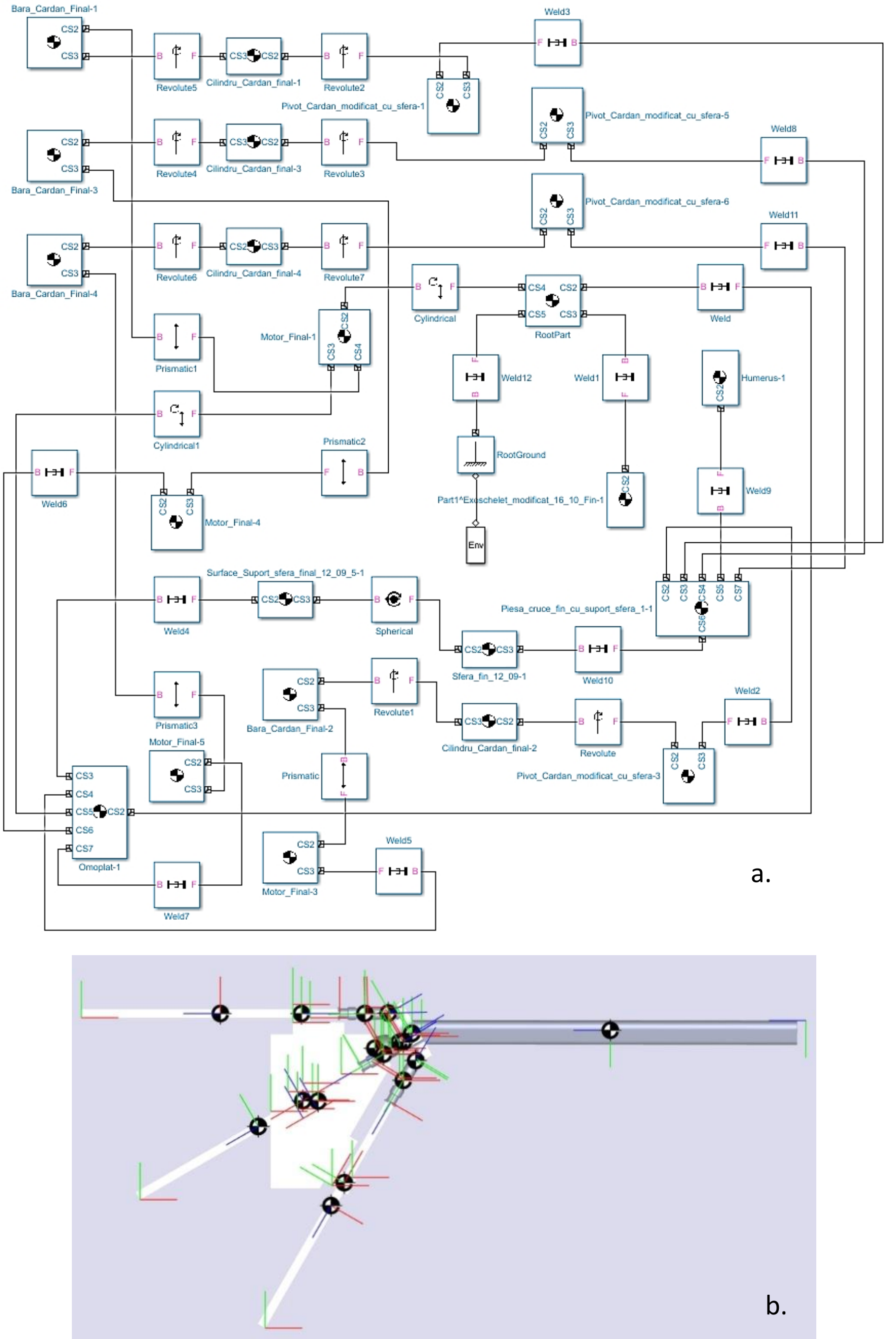

Fig. 3. a. Simulation scheme implemented in Matlab; b. The result of the simulation, with the arm stretched horizontally.

The shoulder blade, in addition to the four "muscles" with which it controls the movements of the Humerus, also supports two other actuating motors, left and right, the correspondents of the "muscle" Biceps (see Fig. 4), with which the forearm can be controlled (see Table 2.).

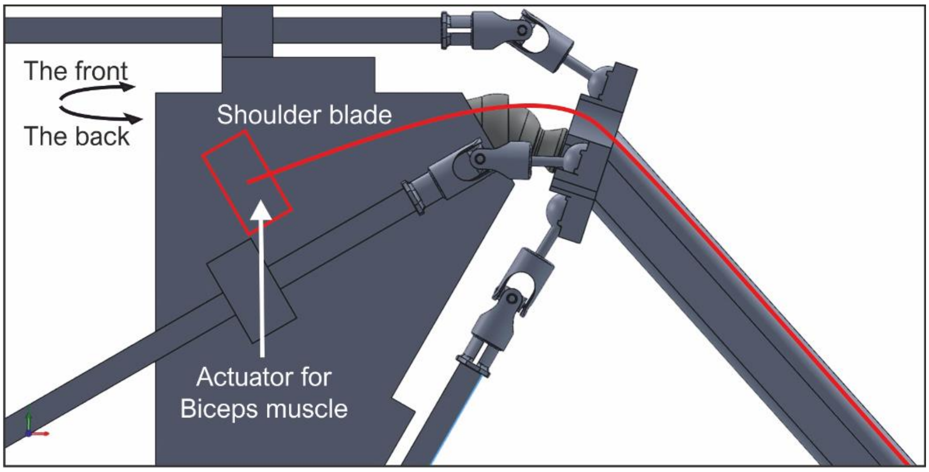

Fig. 4. The actuator motor for the control of the "Biceps" muscle, mounted on the shoulder blade, controls the movements of the forearm.

Table 2. Control and actuation elements

Shoulder blade:
• it has the abstracted form of the human shoulder blade • supports 6 actuator motors (1 up, 1 down, 1 + 1 left, 1 + 1 right; 1 actuator on the left represents the Biceps that connects to Radius, 1 actuator on the right represents the Biceps that connects to Ulna) • supports the hemisphere, the shoulder joint (Glenoid cavity (Glenoid fossa))
Humerus:
• tube type, with triangular section, with one face facing up • supports the pivot that has a sphere in the part from the shoulder blade (the head of the Humerus) • the sphere of the pivot slides inside the hemisphere of the shoulder blade • supports the arm sockets, in the form of ✚, the place where the flexible connections are fixed (the abstracted form of the elevations of the upper part of the Humerus)

"Muscle" Biceps in the natural way is a muscle on two joints, indirectly helping to raise the arm, its main role is to raise the forearm. In the part from the shoulder the muscle is divided in two, the short end, which is connected by the tip of the Coracoid Process, and the long end, connected on the supraglenoid tubercle of the shoulder blade. Because the two tendons are positioned front-to-back against the shoulder wrist, I chose, for the control of the "muscle" Biceps, to position the actuating motors, one on the sides of the shoulder blade. This is part of the further development of the project, when I will also build the elbow.

Naturally, the Biceps muscle is connected to Radius, and to Ulna the Brachialis muscle. In this project, the Biceps "muscle" will consist of two flexible cross-links, meaning that the right motor link will be linked to the left forearm (Ulna side), and the left motor link will be linked to the right forearm. (Radius part). This will lead to the possibility of rotating the forearm with 30 degrees to the left and right, which will be made in one piece, not two, as in the natural way.

Conclusions

Reducing the number of actuator motors needed to control the artificial muscles for moving the elements leads to reduced electricity consumption, weight reduction, the driving algorithm is simpler. The disadvantage is that through simpler movements, much of the fluidity of the movement is lost and also the arm's degree of maneuverability is drastically reduced.

Further, the project will have three directions of development. For the first one, I want to replace the cardan type links with flexible strings; for the second, the elbow wrist and later for the hand and fingers, when I will try to implement the same type of wrist, consisting of a hemispherical socket and the pivot with the sphere, obviously adapted to the specific of the wrists. In the third direction of development, I will try to transform this robotized arm into an exoskeleton type, which will wear a human arm. In this case, the elements of the exoskeleton will be like a medieval armor over the arm, where the socket with the hemisphere and the pivot with the sphere will be even the wrist of the natural human arm.

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